Government of Karnataka

Department of Technical Education

Board of Technical Examinations

Course: Thinking Programming with Python **Course Code: 25CS21I**

Time: 3 Hrs **Max Marks: 100**

Instructions: For Part-A questions, only the first written answers will be considered for evaluation.

PART A

I. Select the correct answer from the choices given: 15X1 = 15

1. What is the purpose of decomposition in computational thinking?
   a) To make the problem more difficult
   b) To break a problem into smaller, manageable parts
   c) To find the final answer directly
   d) To memorize all problem details
2. Which component of computational thinking helps in creating step-by-step instructions to solve a problem?
   a) Pattern Recognition b) Abstraction c) Algorithm Design d) Decomposition
3. Which of the following is a sorting algorithm?
   a) Dijkstra's Algorithm b) Bubble Sort c) Linear Search d) Caesar Cipher
4. Which of these is a primitive data type in Python?
   a) list b) int c) dict d) set
5. What is the symbol used for comments in Python?
   a) // b) <!-- --> c) # d) /* */
6. Why do we use comments in programming?
   a) To make the program run faster b) To explain the code and improve readability
   c) To create variables automatically d) To convert data types
7. Which of the following statements is used to take input from the user in Python?
   a) get() b) input() c) scan() d) read()
8. Which keyword is used in Python to exit a loop immediately?
   a) skip b) exit c) break d) stop
9. Which of the following will correctly print the result of 5 multiplied by 3 using string formatting with an f-string?
   a) print("5 * 3 = {5 * 3}") b) print(f"5 * 3 = 5 * 3")
   c) print(f"5 * 3 = {5 * 3}") d) print("5 * 3 = " + 5 * 3)
10. What is a syntactical error in Python?
    a) An error caused by incorrect logic
    b) An error due to incorrect use of Python grammar or rules
    c) An error that happens only at runtime d) An error that the user cannot fix

11. What type of error occurs if you forget a colon : at the end of an if statement?

a) Semantic Error b) Logical Error c) Syntactical Error d) Runtime Error

12. Consider this code snippet:

```
x = 10
y = 0
result = x / y
print("Result is:", result)
```

What type of error will this code produce when run?

a) Syntactical Error b) Semantic Error

c) Runtime Error (ZeroDivisionError) d) No Error

13. Which keyword is used to define a function in Python?

a) function b) def c) func d) define

14. Which file must be present in a folder for it to be treated as a Python package?(R)CO5

a) main.py b) package.py c) __init__.py d) start.py

15. What will be the output of the following code?

```
def greet(name="Guest"):
  print("Hello,", name)
greet()
```

a) Hello, name b) Hello, Guest c) Hello d) Error because no argument was passed

II Fill in the blanks by choosing appropriate answer from those given in the bracket:

(continue, Object-Oriented, Sorting, syntactical, module) **5X1 = 05**

1. _____ is the process of arranging data in a specific order, such as ascending or descending.
2. The programming paradigm that organizes code using objects containing data and methods is called _____ programming.
3. In a while loop, the _____ statement can be used to skip the current iteration and continue with the next iteration of the loop.
4. A _____ error occurs when the Python interpreter cannot understand the code because it violates the rules of the language, such as missing colons or parentheses.
5. A Python file containing functions and variables is called a _____.

PART B

III. Answer any FIVE questions: **5X2 = 10**

1. What is decomposition in computational thinking? Explain with one example.
2. Explain what a searching problem is and give one example.
3. What is syntax in programming? Give one example.
4. Name any two types of conditional statements used in Python.
5. Define a syntactical error in Python.
6. Give one example of a semantic error.

7. Name any two types of import statements used in Python.
8. What is a module in Python?

PART C

IV. Answer any FIVE questions: **5X3 = 15**

1. Describe sorting problems and explain why sorting algorithms are important, with one example.
2. Explain the difference between variables and constants in programming, with examples
3. Why is it important to follow best programming practices? Explain with examples
4. Explain the purpose of the break statement in loops with the help of an example.
5. What is an f-string in Python? How is it different from using the format() method?
6. Differentiate between syntactical errors and semantic errors in Python.
7. What is the difference between a Python module and a package? Explain with an example.
8. Explain how the from module import function statement works with an example.

PART D (Section I)

V. Answer any FIVE questions: **5X5 = 25**

1. Discuss the applications of Computational Thinking in various fields. Give examples to support your answer.
2. You are planning a school event that includes guest speeches, games, and lunch for all students. Apply the steps of Computational Thinking to organize this event effectively.
3. Write a Python code snippet to create two variables, store your name and age in them, and print a sentence introducing yourself.
4. Explain the difference between if-else and if-elif-else statements in Python. Write an example of each to show how they work.
5. Write a Python program that asks the user to enter an integer. If the number is even, print "Even number". If it is odd, print "Odd number".
6. Explain with examples syntactical errors occur in Python.
7. Explain with example semantic errors occur in Python .
8. Create a Python module named greetings.py that contains a function say_hello() which prints "Hello, welcome!". Then, write a script to import this module and call the function.

PART D (Section II)

VI Answer any THREE questions: **10X3 = 30**

1. A teacher wants to create a timetable for five classes with different subjects and teachers. Apply the steps of Computational Thinking to plan how you would solve this problem.
2. Explain the structure and working of a for loop and a while loop in Python. Write examples to illustrate each case.
3. Describe different methods of formatting output in Python. Explain string concatenation,

format() method, and f-strings with examples showing how each method works.

4. The following program runs without syntax errors, but the output is incorrect due to a semantic error. Identify the logical mistake, explain it, and rewrite the correct program.

```
# Program to calculate the average of two numbers
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
average = num1 + num2 / 2
print("The average is:", average)
```

5. Define a function assign_priority() for ticketing system which assigns priorities to support tickets based on the issue type (low, medium, high and returns a priority level.

****